RISHABH ANUP

Unity Game Developer
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PROFILE

Motivated and detail-oriented MSc graduate with hands-on experience in Unity game development and AI integration. Skilled in C#, single-player and multiplayer systems, and gameplay mechanics. Eager to begin a career as a Game Programmer, contributing to engaging and innovative projects while expanding professional skills.

TECHNICAL SKILLS

Engine & Language: Unity (2D/3D), C#, WebGL

Gameplay Systems: State machines, player controllers, combat systems, boss AI, grid-based movement, object pooling, ScriptableObject-driven architecture, event-based decoupling

Multiplayer: Photon PUN2 — real-time synchronisation, RPCs, leaderboards, match timers

AI: Unity ML-Agents, enemy AI, pathfinding, behaviour design

UI Programming: Unity UI, TextMeshPro, health bars, timers, score systems, HUD design

XR Development: Meta Quest 2, Meta Quest Pro, HoloLens 2, MRTK, SteamVR, AR/VR interaction

Tools & Workflow: Cinemachine, Tilemap, ScriptableObjects, Git, GitHub, itch.io, Google Play

Art & Animation: Sprite integration, Animator controllers, blend trees, particle systems

GAME PROJECTS

Auto Shooter Roguelite *- Android/WebGL - Unity, C#, URP (In Development) [itch.io]*

- Building a mobile-first survivor roguelite for Android with auto-firing weapons, enemy waves, and a deep upgrade system.
- Architected a ScriptableObject-driven weapon system with 4 weapon types (Magic Bullet, Orbit Orbs, Frost Nova, Chain Lightning), each with 3 upgrade tiers and planned evolutions.
- Implemented a weighted random upgrade card system with 4 rarity tiers (Common - Legendary) influenced by a Luck stat. level-up pauses gameplay and presents 3 contextual cards.
- Designed a timeline-based WaveManager using WaveData ScriptableObjects, introducing 4 enemy types progressively over a 10-minute run with stat scaling, pack spawns, and a boss trigger.
- Built 4 enemy archetypes: Walker (baseline), SwarmBug (pack threat), Bloater (proximity explosion with warning indicator and chain detonation), Charger (state machine)
- Engineered a full object pooling system for bullets, enemies, and XP gems.
- Developed a static GameEvents hub for fully decoupled communication across health, XP, upgrades, UI, and wave systems.
- Set up Cinemachine 3.x with PositionalComposer, lookahead, and CinemachineImpulseSource for context-sensitive screen shake. (player hit, Bloater explosion, Charger impact)
- Implemented UI systems: smooth health bar with colour shift, XP bar with level text, run timer, upgrade card UI with rarity colouring, game over and victory overlays with fade and run stats.

Shutdown Protocol *- 2D Boss Rush - Unity, C#, WebGL [itch.io]*

- Developed a complete 2D side-view boss rush.
- Engineered three mechanically distinct bosses using enum-based state machines with multi-phase behaviour and attack telegraphing.
- Built reusable boss architecture via an abstract BossBase class.
- Implemented a complete player system including variable-gravity jump, melee hitbox with directional up-strike, dash with i-frames, and modular health with invincibility frames.
- Developed a sequential boss flow manager handling dynamic boss spawning, inter-boss player healing, and cross-scene UI rebinding.

- Integrated Cinemachine screen shake, hitstop freeze frames, and particle hit effects for game feel; shipped as WebGL on itch.io with per-boss music crossfading.

Mutation Wars *- Multiplayer Strategy - Unity, Photon PUN2, ML-Agents*

- Designed and developed a real-time multiplayer battle game with Rock-Paper-Scissors form mutation dynamic.
- Architected a state machine system managing player form transitions, ability activation, and speed variations across networked clients.
- Integrated Photon PUN2 for player synchronisation, tagging system, live leaderboard, and match timer.
- Began integration of AI opponents for singleplayer mode using Unity ML-Agents training pipeline.
- Implemented UI systems including countdown timers, power-up icons, and strategic alert notifications.

Tower Defense Roguelike *- Unity, C#*

- Built a wave-based tower defense game with roguelike upgrade paths, XP progression, and scalable difficulty curves.
- Implemented modular combat architecture using projectile inheritance and event-driven systems via C# Actions.
- Designed an XP and levelling system using ScriptableObjects with exponential XP scaling and level-up handling.

Grid-Based Dungeon Crawler Demo — *Unity, C#*

- Building a turn-based 2D dungeon crawler with grid-based movement, tile collision via Unity Tilemap, and a discrete turn system.
- Implemented three enemy archetypes with distinct AI behaviours: Skeleton (step-toward pathfinding), Ghost (adjacency teleport with fade effect), and Skeleton Knight. (turn-skipping heavy attacker)

EMPLOYMENT HISTORY

Unity Developer | Travancore Analytics | Aug 2022 – Jan 2024 | Kochi, India

- Developed interactive AR/VR applications for Meta Quest 2, Meta Quest Pro, and HoloLens 2 using Unity.
- Built a VR forklift training simulator with realistic driving mechanics, haptic feedback, and object interaction.
- Ported the simulator from SteamVR (HTC Vive) to standalone VR (Meta Quest 2), ensuring optimisation and performance improvements.
- Designed and implemented login and authentication systems, integrating APIs to manage user sessions and interactions.
- Contributed to TwinVerse, a collaborative mixed-reality platform; implemented avatar/scene selection, login handling, and QR scanning for dynamic model loading using MRTK.

EDUCATION

MSc Data Science and Artificial Intelligence | University of Liverpool | Sept 2024 – Sept 2025 | Liverpool, UK

B.Tech Electronics and Communication Engineering | Sree Chitra Thirunal College of Engineering | Aug 2018 – Sept 2022 | Trivandrum, India